

An Agmon bound for the Kogut–Susskind ground state: large fields are suppressed at rate $1/g^2$ per plaquette

The estimate that the flow-before-decimation note identified as the missing ingredient exists and is standard: Agmon’s method applies to the Kogut–Susskind Hamiltonian because its configuration space is a compact Riemannian manifold, its magnetic term is a bounded potential and its ground state is positive. Writing

$$H = \frac{\hbar c}{a} \left[\underbrace{\frac{g^2}{2}}_{A'} (-\Delta) + \underbrace{\frac{2}{g^2}}_{B'} V \right], \quad V(U) = \sum_p (N - \text{Re tr } U_p),$$

the identity

$$A' \int |\nabla(e^\rho \Omega)|^2 + \int (B'V - e_0 - A'|\nabla \rho|^2) e^{2\rho} \Omega^2 = 0, \quad e_0 = \frac{aE_0}{\hbar c},$$

holds for every Lipschitz ρ and gives exponential decay of the ground state into the region where the magnetic energy exceeds its mean. The Agmon metric is $\sqrt{(B'V - e_0)/A'} |dU|$, with $B'/A' = 4/g^4$, and a configuration carrying an excess of n excited plaquettes sits at Agmon distance of order $2n\sqrt{v}/g^2$ from the allowed region, where v is the excess per plaquette. Hence

$$|\Omega(U)|^2 \lesssim \exp \left[- \frac{cn\sqrt{N}}{g^2} \right],$$

the ground-state measure suppressing large-field regions at a rate $\propto 1/g^2$ per plaquette, **the same coupling dependence as the Euclidean Wilson weight** e^{-S_w} with $\beta = 2N/g^2$. The suppression is strong exactly at weak coupling, which is where the renormalization steps of the route live, and weak at strong coupling, where the cluster expansion already works. This supplies the Hamiltonian counterpart of the large-field estimate whose absence the typical-field note recorded, and it does so by a standard technique rather than a new one. Constants explicit; the distance estimate is a scaling argument and is labelled as such; nothing promoted.

1. Setting and the ground-state identity

The configuration space $\mathcal{M} = G^{\mathcal{E}}$ is a compact Riemannian manifold with the product bi-invariant metric, of dimension $|\mathcal{E}| \dim G$, and $-\Delta$ is its Laplace–Beltrami operator. By T1 of the obligations map the ground state Ω is strictly positive and satisfies

$$-A' \Delta \Omega + (B'V - e_0) \Omega = 0, \quad A' = \frac{g^2}{2}, \quad B' = \frac{2}{g^2}, \quad e_0 = \frac{aE_0}{\hbar c},$$

after dividing by $\hbar c/a$.

Lemma 1 (Agmon identity). For every Lipschitz $\rho : \mathcal{M} \rightarrow \mathbb{R}$,

$$A' \int_{\mathcal{M}} |\nabla(e^\rho \Omega)|^2 d\mu + \int_{\mathcal{M}} (B'V - e_0 - A'|\nabla \rho|^2) e^{2\rho} \Omega^2 d\mu = 0.$$

Proof. Multiply the eigenvalue equation by $e^{2\rho} \Omega$ and integrate; there is no boundary term because $\mathcal{M}$ is closed. Using $\int (-\Delta \Omega) e^{2\rho} \Omega = \int \nabla \Omega \cdot \nabla(e^{2\rho} \Omega)$ and the pointwise identity

$$|\nabla(e^\rho \Omega)|^2 = e^{2\rho} |\nabla \Omega|^2 + 2e^{2\rho} \Omega \nabla \rho \cdot \nabla \Omega + e^{2\rho} \Omega^2 |\nabla \rho|^2 = \nabla \Omega \cdot \nabla(e^{2\rho} \Omega) + e^{2\rho} \Omega^2 |\nabla \rho|^2,$$

the first term becomes $\int |\nabla(e^\rho \Omega)|^2 - \int e^{2\rho} \Omega^2 |\nabla \rho|^2$. Adding $\int (B'V - e_0) e^{2\rho} \Omega^2$ gives the identity.

□

Corollary 2 (Agmon estimate). Let $\Sigma_\epsilon = \{U : B'V(U) - e_0 \geq \epsilon\}$ and let ρ be Lipschitz with $A'|\nabla\rho|^2 \leq B'V - e_0 - \epsilon/2$ on Σ_ϵ and $\rho \leq \rho_{\max}$ on $\mathcal{M} \setminus \Sigma_\epsilon$. Then

$$\frac{\epsilon}{2} \int_{\Sigma_\epsilon} e^{2\rho} \Omega^2 d\mu \leq (e_0 + A' \|\nabla\rho\|_\infty^2) e^{2\rho_{\max}} \int_{\mathcal{M} \setminus \Sigma_\epsilon} \Omega^2 d\mu \leq C e^{2\rho_{\max}}.$$

Proof. Drop the nonnegative gradient term in Lemma 1 and split the integral over Σ_ϵ and its complement, where the integrand is bounded below by $-(e_0 + A' \|\nabla\rho\|_\infty^2)$. $\square$

The admissible weights are exactly the functions with $|\nabla\rho| \leq \sqrt{(B'V - e_0)/A'}$, so the largest is the **Agmon distance**

$$d(U) = \inf_{\gamma: U \rightarrow \{B'V \leq e_0\}} \int_\gamma \sqrt{\frac{B'V - e_0}{A'}} |d\gamma|,$$

and Corollary 2 with $\rho = (1 - \delta)d$ gives $\int e^{2(1-\delta)d} \Omega^2 \leq C(\delta)$.

2. The Agmon distance of a large-field region

Write $\bar{v} = e_0/B'$ for the mean magnetic energy, which satisfies $\bar{v} \leq \langle V \rangle_{\text{Haar}} = N|\mathcal{P}|$ because $E_0 \leq \langle 1, H1 \rangle$ and $\langle \text{Re tr } U_p \rangle_{\text{Haar}} = 0$. Consider a configuration U whose magnetic energy exceeds $\bar{v}$ by an amount carried on n plaquettes, each in excess by $v = O(N)$, so $V(U) - \bar{v} \simeq nv$.

Flat length. Relaxing those n plaquettes requires moving of order n link variables by an angle of order one. In the product metric the flat distance is of order $\sqrt{n}$, since n coordinates each move $O(1)$.

Agmon weight. Along such a path the excess decreases from nv to 0, with mean of order $nv/2$, so the weight $\sqrt{(B'V - e_0)/A'} = \sqrt{(B'/A')(V - \bar{v})}$ is of order $\sqrt{(4/g^4)nv/2} = \sqrt{2nv}/g^2$.

Distance. Multiplying, and keeping only the order,

$$d(U) \simeq \frac{\sqrt{2nv}}{g^2} \cdot \sqrt{n} = \frac{n\sqrt{2v}}{g^2}.$$

Proposition 3 (scaling form). With the distance estimate above, Corollary 2 gives, for the ground-state measure,

$$\int_{\{n \text{ excess plaquettes}\}} \Omega^2 d\mu \lesssim C \exp \left[- \frac{2(1-\delta)\sqrt{2v}}{g^2} n \right],$$

exponential in the number of excited plaquettes with rate $\lambda = 2\sqrt{2v}/g^2$ per plaquette, $v = O(N)$.

The flat-length and mean-excess steps are scaling estimates; the identity of Lemma 1 and the estimate of Corollary 2 are exact.

3. Comparison with the Euclidean weight

The Euclidean Wilson measure is $e^{-S_w} d\mu$ with

$$S_w = \frac{2N}{g_E^2} \sum_p \left(1 - \frac{1}{N} \text{Re tr } U_p \right) = \frac{2}{g_E^2} V,$$

suppressing a configuration with n excited plaquettes by e^{-2nv/g_E^2} . Proposition 3 gives $e^{-2n\sqrt{2v}/g^2}$ for the Hamiltonian ground-state measure. The two agree in their dependence on the coupling, $1/g^2$ per excited plaquette, and differ in the power of the excess per plaquette, linear against square-root, which is the usual difference between a Boltzmann weight and a WKB weight: the Agmon bound is a tunnelling estimate and the Wilson weight is a direct cost.

The direction is the useful one. The suppression rate grows as $g \rightarrow 0$, so the large-field regions that the typical-field note could not control in the Hamiltonian framework are exponentially rare in the ground-state measure precisely at weak coupling, which is where the renormalization steps of the position note §6 are needed. At strong coupling the rate degrades, and there the cluster expansion of the T2 note already applies.

4. What this supplies and what it still lacks

Supplies. A Hamiltonian counterpart of the large-field estimate, by a standard technique, with an explicit rate. It converts the state-relative criterion demanded by the flow-before-decimation note §3 into a computable statement: expectations of quantities that grow exponentially in the local field strength are finite in the ground state as long as their growth rate stays below $\lambda = 2\sqrt{2v}/g^2$.

Lacks. Three things. The estimate is for the ground state, and the route needs it for the whole low-energy subspace, which requires either the same argument for low-lying eigenfunctions, available with e_0 replaced by e_1 at the cost of a smaller forbidden region, or a spectral-projection version. The distance estimate of Section 2 is a scaling argument and needs a proof with constants, which is a geometric computation on $G^\mathcal{E}$. And the resulting suppression must be matched against the growth $e^{2t\|G\|_\infty}$ of the flow Jacobian: by the lattice-truncation note that factor is a pure number per doubling, so the match is comfortable within one step and the question is again the composition of steps.

5. Consequence for STATE

The missing ingredient of the previous note has a standard source. The Kogut–Susskind ground state obeys an Agmon bound whose rate, $\lambda \simeq 2\sqrt{2v}/g^2$ per excess plaquette, reproduces the coupling dependence of the Euclidean Wilson weight and is strong exactly at weak coupling. The next items, in order of tractability: prove the distance estimate of Section 2 with constants; extend the bound from the ground state to the spectral subspace below the gap; then assemble the state-relative decimation criterion that these two make possible, which is the first version of the renormalization step in which the difficulty is located in the composition of steps rather than in any single one.